



Spatiotemporal Changes and Predictive Modeling Regarding Carbon Emissions in the Yellow River Basin: Integrated Method Incorporating Tapio Decoupling Model and GM(1,1) Grey System Theory

Hao Huang^{1,2*}, Yiwei Huang¹, Bowen Zhang¹, Wenxuan Duan¹, Yujun Hu¹, Jing Wang¹, Shihang Cheng¹, Anxue Li^{1,2}, Bojing Zhang³

¹ School of Economics and Management, Shandong Institute of Petroleum and Chemical Technology, 257000 Dongying, China

² Key Laboratory of Digital Transformation and Application of Energy and Chemical Enterprises, 257000 Dongying, China

³ School of Finance, Changchun University of Finance and Economics, 130000 Changchun, China

* Correspondence: Hao Huang (13805460321@139.com)

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Abstract: The Yellow River Basin plays a crucial part in China's plan to meet the "Dual Carbon" targets since reduction of carbon emissions has a profound impact on the national path towards low-carbon development. This research adopted the Tapio decoupling model and the GM(1,1) grey forecasting model to examine how carbon emissions interacted among nine provinces and regions within the Basin from 2006 to 2021. It also predicted future emission tendencies and obtained the following key results: (1) From 2006 to 2021, total carbon emissions in the Yellow River Basin grew at an average annual rate of 4.17%, with the upper and middle reaches driving most of the increase while the lower reaches continued to account for the majority of emissions; (2) The decoupling relationship between economic growth and carbon emissions demonstrated different patterns over space and time. Mostly the upper areas had a type of expansive negative decoupling, the middle areas fluctuated extensively, and the lower areas mostly encountered a weak decoupling situation. During the study period, there was a shift from extensive growth to more differentiation; and (3) According to the projections of the GM(1,1) grey forecasting model, carbon emissions in these nine provinces along the Yellow River would keep increasing but at a slower pace. Therefore, the paper put forward targeted suggestions including promoting industrial green transformation in different regions, developing inter-regional cooperation mechanisms for emission reduction, and optimizing the basin-wide energy system so as to promote sustainable, green, and low-carbon development in the Yellow River Basin.

Keywords: Yellow River Basin; Dual Carbon; Emission reduction; Tapio decoupling model; Grey prediction model

1 Introduction

Climate change represents an imperative global challenge. As the world's most populous developing country and a major source of carbon emissions, China bears a critical responsibility to address climate change and decrease its carbon emissions. In September 2020, President Xi pledged at the United Nations General Assembly that China would have reached a peak in carbon dioxide emissions by 2030 and achieved carbon neutrality by 2060 [1]. This strategic objective, known as the 3060 Dual Carbon target, outlines the trajectory and policy framework for China's initiative to lower carbon emissions.

The Yellow River Basin is of great significance in China as it serves as a crucial ecological buffer, a major economic center, and a main supplier of energy resources. The Yellow River Basin possesses over 95% of China's rare earth reserves, more than 45% of its coal and non-ferrous metal mineral reserves, and 26% of its petroleum reserves, making it the core energy-producing basin in China [2]. It is also a key industrial agglomeration area in China, dominated by high-pollution, high-water-consumption and high-energy-consumption industries such as coal chemical, steel and non-ferrous metal smelting. Existing studies have confirmed that these industrial activities not

only directly generate massive carbon emissions, but also produce significant inter-provincial spatial spillover effects through industrial chains and regional trade, making the basin a major contributor to national carbon emissions [3].

Regions adjacent to the Yellow River Basin have demonstrated marked advancement in low-carbon progression, carefully adhering to the national goals for reaching carbon-neutral status and peaking carbon dioxide emissions as outlined in the State Council's 2021 Action Plan for Carbon Peaking before 2030. In particular, the upper reaches covering Qinghai, Gansu, Ningxia and Inner Mongolia have been designated as the national core area for integrated hydro-solar-wind-storage clean energy bases under the 14th Five-Year Plan, boasting abundant complementary renewable energy resources: hydropower resources are concentrated in Qinghai with cascading hydropower stations providing flexible peak regulation capacity, while solar and wind resources are widely distributed across the four provinces with annual shortwave radiation reaching 1700–2200 kWh/m² and average wind speed exceeding 2.5 m/s in most areas [4]. Kong et al. [5] researched and assessed the current state of soil erosion restoration in the Yellow River Basin. Compared with data from the 1990s, monitoring data from 2019 showed a 43% reduction in the area affected by soil erosion, with a decrease of 80% in areas experiencing severe erosion or higher. The degree and intensity of soil erosion have both been diminished in a “double reduction”, indicative of substantial progress in ecological restoration. Despite these developments, inconsistencies in strategic planning and development priorities exist within the provinces and areas of the Yellow River Basin, with variations in economic growth and low-carbon goals observed across the upper, middle, and lower reaches at different intervals. Consequently, regional and temporal discrepancies necessitate meticulous consideration. Customizing emission reduction tactics to the historical context, industrial structure, and economic profile of each province and basin is imperative. An approach grounded in the practical realities of each area should be implemented to foster uniform emission reduction and low-carbon advancement across all provinces and regions. In 2021, General Secretary Xi Jinping emphasized the need to establish the “four beams and eight pillars” for the protection and governance of the Yellow River, address ecological and environmental problems, advance ecological protection and restoration, improve the governance system, and make new progress in high-quality development, ushering in a new era of ecological conservation and high-quality development in the Yellow River Basin [6]. To encourage coordinated emission reduction initiatives within the Yellow River Basin, a mission of significance to the country's pursuit of its concurrent carbon goals, a precise assessment of the present state, and projected growth of carbon dioxide releases in the Basin are essential.

This study analyzed the trends of carbon emission evolution and evaluated the decoupling degree between carbon emissions and economic growth across the nine regions, with an aim to clarify the spatiotemporal characteristics of carbon emissions in the Yellow River Basin and provide scientific predictions for future carbon emission pathways in the region.

2 Literature Review

Carbon emissions have long been a focal area of academic research. Existing studies have advanced along three core dimensions. Carbon emission accounting systems, driving mechanism identification, and trend prediction models, thus providing crucial theoretical support and empirical evidence for the formulation of low-carbon development policies across countries.

Carbon emission accounting is the foundation of all low-carbon research. Following the establishment of the Intergovernmental Panel on Climate Change (IPCC) in 1988, efforts were initiated to develop a unified global standard for carbon emission accounting. In 1996, the IPCC published the IPCC Guidelines for National Greenhouse Gas Inventories, which for the first time systematically specified the accounting scope, sectoral classification, and coefficient selection methods for greenhouse gas emissions. This document established the standard framework for national-level carbon emission accounting worldwide and remained the core reference for official statistics and academic research in various countries [7]. Building on this foundation, scholars began to explore refined accounting methods at the sectoral level. Du et al. [8] constructed a carbon emission accounting system for the cement industry in China from both supply and demand perspectives. By integrating cement production and consumption data from 31 provinces across the country, they systematically distinguished between direct and embodied emissions in cement production for the first time, quantified the carbon emission transfer effects caused by inter-regional cement trade, and revealed the long-term impact of demand-side driving factors on carbon emissions of the cement industry. Zhang et al. [9] creatively applied the concept of “hub” to carbon flow tracing and carbon accounting, extending the conventional energy hub to a material flow-based energy hub. On this basis, they proposed an integrated material-energy-carbon hub model to optimize carbon accounting methods and explore the impact of material and changes of energy flow on carbon dioxide emissions during steel production.

Identifying the driving factors of carbon emissions is a prerequisite for formulating emission reduction policies, and its methodological system has evolved from simple index decomposition to complex model decomposition. In 1989, Japanese scholar Kaya proposed the Kaya identity, which for the first time decomposed carbon emissions into four core factors: population, total economic output, energy intensity, and energy structure, providing the most fundamental analytical framework for driving factor research [10]. However, traditional index decomposition

methods have the inherent defect of ineliminable residual terms. In 2015, Ang et al. published the implementation guide for the Logarithmic Mean Divisia Index (LMDI) decomposition method, which systematically summarized the theoretical basis and application specifications of the LMDI method. Due to its residual-free and stable results, it has become the most widely used decomposition method at present [11]. Since the beginning of the 21st century, research on driving factors has further expanded to spatial and dynamic dimensions. Chang et al. [12] extended the Kaya identity to identify carbon emission influencing factors and quantified their contribution degrees using the LMDI method, establishing a quantitative relationship model for carbon emission prediction. Chen et al. [13] applied the synthetic control method to empirically evaluate the governance effect of China's sulfur dioxide emissions trading pilot scheme (SETPS), confirming that market-based environmental regulation tools can significantly reduce industrial pollution emission intensity.

Grasping the trend of carbon emission development and predicting their future trajectories helps identify emission reduction gaps, optimize policy recommendations, and bridge scientific cognition and policy action, which is of great significance for low-carbon emission reduction. After half a century of academic evolution, the carbon emission prediction method system is gradually improved, going through four developmental stages: The foundation of traditional statistical models, the expansion of scenario analysis models, the breakthrough of decomposition-integration models, and the innovation of machine learning models. The Autoregressive Integrated Moving Average (ARIMA) model proposed by Box and Jenkins in 1970 was the earliest time series model applied to carbon emission prediction, suitable for short-term stationary sequence prediction [14]. This method converts non-stationary data into stationary data through difference processing and can effectively capture short-term linear change laws. Since the 1980s, the ARIMA model has been gradually introduced into the energy and carbon emission fields and has become a standard tool for national-level short-term carbon emission prediction after 2000. The model has been widely used in the prediction research of large sample data such as energy consumption and industrial emissions. Sharma et al. [15] applied the ARIMA time series model to forecast carbon emissions in India using 41 years of data (1980–2021), verifying that the model can effectively capture the trend and seasonality of carbon emission sequences. However, such methods have always been unable to get rid of the constraint arising from the assumption of large sample stationary data. It is difficult to characterize the nonlinear fluctuation characteristics of carbon emission sequences caused by policy shocks and technological breakthroughs, resulting in poor applicability in regional studies with severe data missing. To address the demand for long-term policy simulation, scenario analysis model emerged in the 1990s to quantify the long-term trend of carbon emissions by setting multi-dimensional development scenarios. The IPCC has taken it as the core method for global prediction since the Third Assessment Report. Zhang and Luo [16] constructed a Long-range Energy Alternatives Planning (LEAP) model to simulate the carbon emission peak path of the construction sector in China, providing a basis for the formulation of industrial Dual Carbon goals. Nevertheless, the prediction results were highly dependent on scenario settings and had strongly subjective uncertainty. To overcome this nonlinear prediction challenge, in 1998 Huang et al. [17] proposed the Empirical Mode Decomposition (EMD) method, which could adaptively decompose complex non-stationary sequences into multiple simple stationary intrinsic mode functions, thus providing a new idea for processing time series with strong volatility such as carbon emissions. After 2010, the idea of “decompose first, predict later, integrate finally” based on EMD has become a research hotspot. Sun and Ren [18] first applied the improved Ensemble Empirical Mode Decomposition (EEMD) to carbon emission prediction, combining it with the Partial Autocorrelation Function (PACF) and Particle Swarm Optimization Back Propagation (PSOBP) model to significantly improve prediction accuracy. Yet this method was computationally complex and the decomposition results were random, rendering it difficult to be promoted at the grassroots level. In recent years, machine learning has become a new direction with its powerful nonlinear fitting ability, and various neural network models have been widely used since 2015. Mardani et al. [19] systematically compared and verified the advantages of machine learning models in predicting long-period and strong-fluctuation sequences, but they suffered from the “black box” characteristics, leading to poor interpretability and the demand for a large amount of high-quality training data. Consequently, they were difficult to be applied in data-scarce regions. All existing methods perform well in their respective applicable scenarios but they still have inherent defects, especially in small sample and poor information scenarios. It is difficult to balance prediction accuracy and interpretability simultaneously. This problem is particularly prominent in regions such as the Yellow River Basin where statistical data of some provinces are incomplete. It has become a key bottleneck restricting accurate regional carbon emission prediction and scientific decision making.

Recently, academic studies have given top priority to looking into carbon emissions in the Yellow River Basin, to highlight its importance in the academic field. Yuan et al. [20] adopted input-output and multi-regional input-output approaches to measure carbon footprints in nine provinces within the Basin, showing cross-provincial and cross-sectoral movements of embedded carbon emissions. Moreover, they incorporated social network analysis to single out key industries causing emissions in the main sectors. Meanwhile, Zhao et al. [21] used quadratic assignment process regression to figure out what influenced carbon emissions, in order to concentrate on regional differences

and create a machine learning algorithm to predict future emissions in that area. Meng et al. [22] applied the LMDI model to decompose the driving factors of land-use carbon emissions in nine provinces of the Yellow River Basin from 1990 to 2018, confirming that economic scale was the primary promoting factor while energy use efficiency was the main inhibitory factor. Liu et al. [23] further focused on the carbon emissions from cultivated land use in the Yellow River Basin, constructed a spatial correlation network using a modified gravity model and social network analysis, and identified that straw burning (over 60%) was the largest source of cultivated land carbon emissions. Wang et al. [24] introduced remote sensing data to conduct a multi-scale (whole basin-sub basin-city) analysis of land-use carbon emissions in 69 cities of the Yellow River Basin from 1990 to 2020, and used the Tapio decoupling model to reveal the long-term decoupling relationship between carbon emissions and economic development.

In summary, existing literature provided important references for research on low-carbon development in the Yellow River Basin, but still there are three main limitations. Spatially, most studies focused on rough comparisons of single provinces or upper-middle-lower reaches, thus lacking systematic analysis of the spatiotemporal evolution of carbon emissions and their decoupling relationship with economic growth across all nine provinces and autonomous regions. Temporally, most studies stopped at historical status analysis, with insufficient forward-looking predictions of future evolutionary trends of carbon emissions. In terms of policy, most studies put forward generic recommendations, whose pertinence and operability need to be improved.

Based on the above research gaps, the marginal contributions of this study were mainly reflected in the following three aspects: (1) In the spatial dimension, this study broke through the limitations of existing zonal studies, took the nine provinces and autonomous regions as a complete research unit, systematically depicted the characteristics of spatiotemporal evolution of carbon emissions, and evaluated the decoupling degree between carbon emissions and economic growth, so as to comprehensively reveal the unbalanced developmental pattern of carbon emissions in the basin; (2) In the temporal dimension, based on the analysis of historical evolution and decoupling relationship, this study used the GM(1,1) grey forecasting model to predict the carbon emission developmental path of the Yellow River Basin from 2022 to 2035, thus providing a scientific basis for the decomposition of medium- and long-term targets of carbon emission reduction in the basin; and (3) In the policy dimension, this study abandoned the one-size-fits-all generic recommendations and proposed differentiated low-carbon developmental strategies combined with the resource endowments and developmental stages of each region, thus significantly enhancing the pertinence and operability of the policies. This study enriched the empirical research results on low-carbon development in the Yellow River Basin and provided decision-making references for the implementation of the ecological protection and high-quality developmental strategy of the Yellow River Basin.

3 Model Construction and Data Description

3.1 Research Methods

3.1.1 Tapio decoupling model

Decoupling, a concept that originated in the realm of physics, is the process by which the connection between two or more entities is diminished or broken [25]. Nowadays, there are two main analytical approaches for decoupling, namely the Decoupling Index Model put forward by the Organization for Economic Cooperation and Development (OECD) depending on initial and final values [26] and the Decoupling State Analysis Model created by Tapio, which centered on changes in the elasticity of growth [27]. The OECD describes decoupling as the separation of economic growth from environmental pollution, whether in terms of different growth rates or absolutely speaking. Relative decoupling occurs when economic growth outpaces that of energy consumption and absolute decoupling is seen when economic growth coincides with either zero or negative growth in energy consumption [28]. The Tapio decoupling model was developed as an improvement and refinement of the OECD decoupling model. By introducing the concept of elasticity and calculating decoupling on a yearly basis without a fixed base year, the model effectively avoids the sensitivity bias inherent in the OECD approach, thus enabling a dynamic analysis. Furthermore, it categorizes decoupling into eight states, forming a more robust indicator system than the OECD model, which allows for a more precise depiction of the complex relationship between economic growth and carbon emissions [29]. The use of Tapio's model for looking into carbon emission decoupling has attracted considerable academic attention worldwide. Yasmeen and Tan [30] made use of this model to evaluate Pakistan's decoupling situation regarding environmental destruction, energy consumption, and economic progress by inspecting the path of carbon emissions and spotting influencing elements and strategic suggestions for shifting towards a low-carbon economy. Driha et al. [31] applied the LMDI model and Tapio's decoupling model to study alterations in tourism-related carbon dioxide emissions throughout the European Union from 2008 to 2022, showing the advantages of innovation in technology, spatial arrangement, and sustainable infrastructure construction on the development of low-carbon tourism. Yang et al. [32] combined the Stochastic Impacts by Regression on Population, Affluence, and Technology (STRIPAT) driving factor decomposition framework and Tapio decoupling model to calculate and predict carbon emissions of Chengdu during 2009–2035, verifying that the decoupling relationship between urban CO₂ emissions and economic development would evolve from weak decoupling to ideal strong decoupling before

2025 under conventional development scenarios. Yang et al. [33] adopted the combined framework of LMDI factor decomposition and Tapio decoupling model to explore China's agricultural carbon emission drivers and decoupling evolution characteristics, confirming that economic output was the dominant positive driving factor of agricultural CO₂ emissions, while energy intensity played a critical inhibitory role, and the overall agricultural carbon-economy relationship gradually shifted toward weak decoupling nationwide.

This study set up an analytical framework based on the Tapio decoupling technique to investigate the different ways in which economic growth and carbon emissions became disengaged from one another in nine separate provinces and areas within the Yellow River Basin. The elasticity figure ε was utilized to measure the connection between the speed of alteration in emissions related to carbon and the pace of development in the economy as shown in the equation below:

$$\varepsilon = \frac{(C_t - C_{t-1}) / C_{t-1}}{(G_t - G_{t-1}) / G_{t-1}} = \frac{\Delta C}{\Delta G} \quad (1)$$

where, C_{t-1} and C_t denote the total CO₂ emissions in the base year $t - 1$ and the target year t , respectively, measured in million tons of CO₂ equivalent (Mt CO₂e); G_{t-1} and G_t represent the Gross Domestic Product (GDP) in the base year $t - 1$ and the target year t , respectively, measured in billion yuan at constant prices (base year = 2000). The decoupling state between CO₂ emissions and economic growth was determined by combining the value of the Tapio elasticity coefficient ε with the directional trends of CO₂ emissions and GDP, as illustrated in Figure 1.

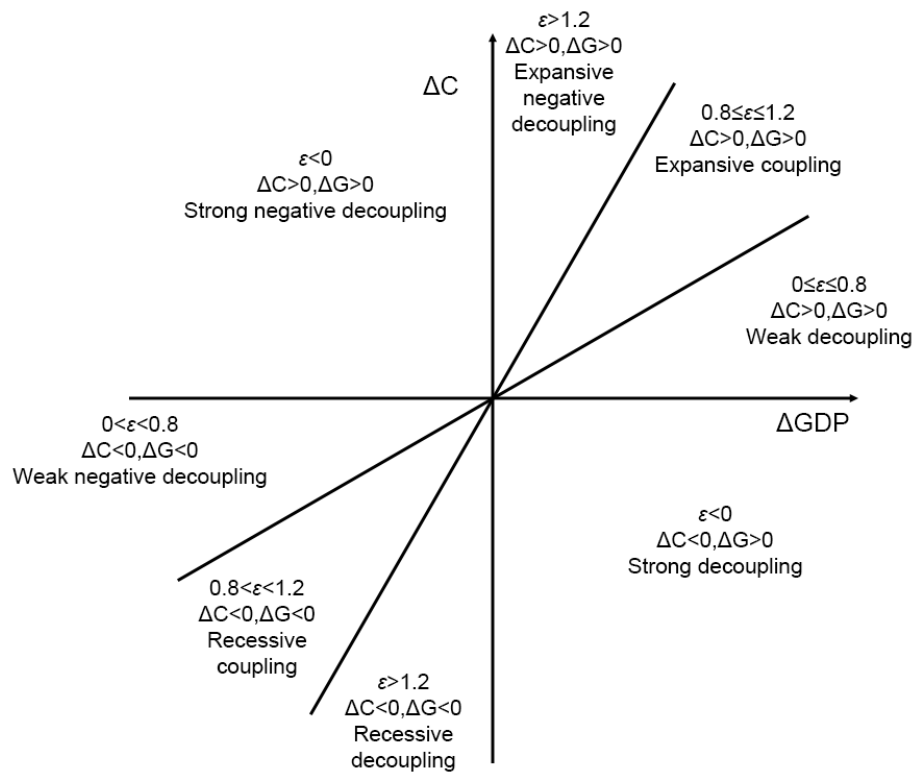


Figure 1. Classification of carbon emission decoupling index

Note: GDP = gross domestic product. ΔC denotes the relative change rate of CO₂ emissions, and ΔG denotes the relative change rate of GDP.

The decoupling state was classified into eight distinct types: strong negative decoupling, expansive negative decoupling, expansive coupling, weak decoupling, strong decoupling, weak negative decoupling, recessive coupling, and recessive decoupling. Specifically, strong decoupling indicates that CO₂ emissions decrease as the economy grows, representing the ideal state and a leading region for low-carbon development. Weak decoupling means CO₂ emissions increase slightly but at a slower rate than economic growth, a relatively ideal state reflecting a high level of low-carbon development. Expansive coupling occurs when the economy and CO₂ emissions grow at roughly equivalent rates with no obvious decoupling trend, an undesirable state indicating a general level of low-carbon development. Expansive negative decoupling refers to economic growth accompanied by fast-growing CO₂ emissions, signifying high-carbon dependence and a poor level of low-carbon development. Strong negative decoupling describes economic recession alongside rising CO₂ emissions, thus imposing severe environmental pressure and constituting the least desirable state for low-carbon development. Weak negative decoupling denotes

simultaneous declines in both the economy and CO₂ emissions, with the economic decline rate exceeding the emissions decline rate. Recessive coupling occurs when the economy and CO₂ emissions decline at roughly equivalent rates and recessive decoupling represents synchronous declines in both variables, with CO₂ emissions decreasing faster than the economy.

3.1.2 GM(1,1) grey forecasting model

In 1982, Professor Deng Julong released a highly influential paper regarding grey systems theory in the well-regarded journal, *Systems & Control Letters*, which signified the start of this theoretical structure. The grey systems forecasting approach was created for systems marked by uncertainty, small sample quantities, and missing information where part of the system details was known but the rest were not explicit. This method aptly utilized accessible information to clarify how the system functioned and what direction it was heading towards, thus illustrating the system's continuous expansion and alteration processes [34]. At the core of grey systems theory lies the grey prediction model, which was considered a crucial element [35]. Among them, the GM(1,1) grey forecasting model, which is a form of this prediction tool, could produce precise predictions with little data and work well even with scarce and faulty information as shown by its wide-spread use in energy research [36]. Huang et al. [37] used system dynamics, Future Land Use Simulation (FLUS) models, and combined three different socio-economic situations to estimate energy demand in the Turpan-Hami Basin up to 2030. Xu et al. [38] utilized the GM(1,1) model to predict carbon emissions in the attempt of South Korea to achieve carbon neutrality. They assessed whether this goal could be reached and demonstrated the effectiveness of the model in carbon emission predictions. Yao et al. [39] used GM(1,1) to predict Suzhou's ecological footprint and proved its superiority over ARIMA for fluctuating environmental data. Wang et al. [40] calculated China's electrical and electronic equipment industrial carbon emissions based on IPCC coefficient method, and applied GM(1,1) grey model to forecast future emission trends; the results showed that the industry's total emissions would reach 136 million tons by 2030 under baseline scenario, and the study proposed EPR mechanism and carbon credit trading paths to cut industrial carbon emissions. Gu et al. [41] utilized GM(1,1) based on forest inventory data to simulate and forecast natural forest carbon storage, confirming the reliable prediction accuracy of grey model for forest carbon sink research.

The present study utilized carbon emission figures from nine provinces across the Yellow River Basin between 2006 and 2021 and adopted the GM(1,1) model for predicting future carbon emissions and their trends from 2022 to 2026. An example of the modeling approach was given and it centered on Qinghai Province as a case study.

The initial sequence for the annual data of carbon emissions from Qinghai $X^{(0)} = \{x^{(0)}(1), x^{(0)}(2), \dots, x^{(0)}(n)\}$, where $x^{(0)}(k)$ represents the original carbon emission value in the k -th year of the study period, and $n = 16$ corresponds to the 16-year sample period from 2006 to 2021.

Specifically, the first-order accumulative generation sequence $X^{(1)} = \{x^{(1)}(1), x^{(1)}(2), \dots, x^{(1)}(n)\}$, which eliminated the randomness of the original data, was established as follows:

$$x^{(1)}(n) = \sum_{k=1}^n x^{(k)}, n = 1, 2, \dots, n \quad (2)$$

where, $x^{(1)}(n)$ denotes the cumulative sum of carbon emissions from the first year to the n -th year.

Construct the differential equation describing

$$\frac{dx^{(1)}(t)}{dt} + ax^{(1)}(t) = b$$

where, a is the development coefficient reflecting the growth trend of the predicted sequence, and b is the grey action quantity representing the change of driving factors in the system.

Use the least square method to solve a, b :

$$\hat{a} = [a, b]^T = [B^T B]^{-1} B^T Y_n$$

$$B = \begin{bmatrix} -\frac{1}{2} [x^{(1)}(1) + x^{(1)}(2)] & 1 \\ -\frac{1}{2} [x^{(1)}(2) + x^{(1)}(3)] & 1 \\ \vdots & \vdots \\ -\frac{1}{2} [x^{(1)}(n-1) + x^{(1)}(n)] & 1 \end{bmatrix}$$

$$Y = \begin{bmatrix} x^{(0)}(2) \\ x^{(0)}(3) \\ \vdots \\ x^{(0)}(n) \end{bmatrix}$$

$x^{(1)}$ of the grey prediction model for:

$$\hat{x}^{(1)}(t) = \left(x^{(1)}(1) - \frac{b}{a} \right) e^{-a(t-1)} + \frac{b}{a} \quad (3)$$

where, t represents the time point, and $x^{(1)}(1) = x^{(0)}(1)$ is the initial value of the accumulative sequence. $x^{(0)}$ of the grey prediction model can be:

$$\hat{x}^{(0)}(t) = \left[\hat{x}^{(1)}(t) - \hat{x}^{(1)}(t-1) \right] \quad (4)$$

where, $\hat{x}^{(0)}(t)$ is the predicted carbon emission value for the t -th year.

3.2 Data Description

This study focused on nine provinces and autonomous regions in the Yellow River Basin from 2006 to 2021. Constant-price GDP data for these regions were obtained from the National Bureau of Statistics of China, and provincial-level total carbon emission data were collected from the Carbon Emission Accounts and Datasets (CEADs). The total carbon emissions adopted in this study thoroughly covered direct CO₂ emissions from two major sources: Fossil fuel combustion (including coal, oil, and natural gas consumption across all sectors) and industrial production processes (including cement, steel, and other key industrial sectors), which strictly follow the accounting framework of the IPCC Guidelines for National Greenhouse Gas Inventories.

4 Results and Analysis

4.1 Basic Situation of Carbon Emissions

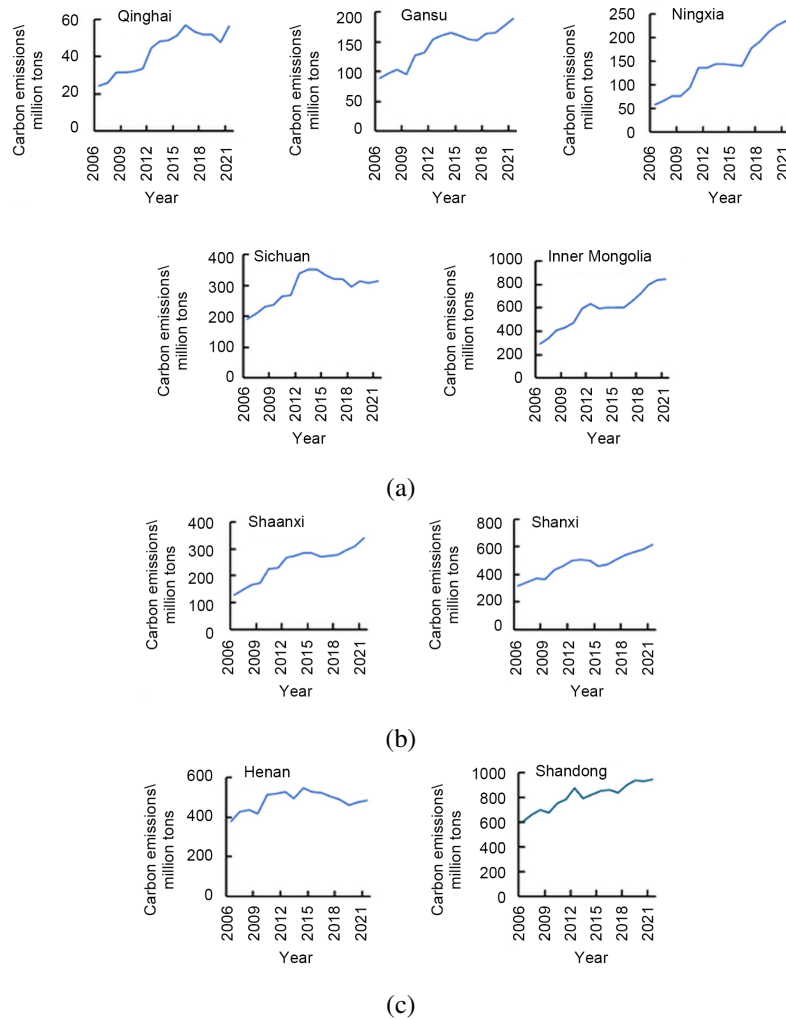
Table 1 displays the carbon output throughout several provinces and autonomous regions of the Yellow River Basin whereas Figure 2 presents the time trend of carbon emissions in the upper, middle, and lower parts of the Basin. When we analyzed Table 1, we could see clear stages of carbon emissions in the Yellow River Basin. From 2006 to 2021, the total carbon emissions increased cumulatively from around 2,086 million metric tons to 4,023 million metric tons; this meant that the average annual growth rate was about 4.17%. It should be noted that Inner Mongolia had a large increase in emissions, going from approximately 291 million metric tons in 2006 to 843 million metric tons in 2021 with an average annual growth rate of around 6.9%. Sichuan, being a very important economic centre, is always rated among the highest in overall carbon emissions. Meanwhile, emissions in Ningxia went up from about 59 million metric tons to 235 million metric tons by 2021, with an average annual growth rate of 9%. These growth rates were higher than those of the Basin, thus showing that heavy industry developed faster, more energy was consumed, and greater emission pressure existed. It was therefore difficult to shift towards low-carbon economies. In Qinghai, the growth rate of total carbon emissions was not too high but there were obvious ups and downs, such as emissions rising sharply by 11.76% from 2015 to 2016. This indicated that the renewable energy sector was expanding even though the region was under ecological protection; however, Qinghai still depended on traditional energy sources for its renewable energy infrastructure and energy-intensive industries, leading to unstable low-carbon progress. Shanxi Province, famous for producing coal, relied heavily on traditional carbon and energy sources but its average annual growth rate slowed down, perhaps because of energy efficiency and emission reduction policies advocated in the 13th Five-Year Plan. Shaanxi Province had an average annual growth rate of about 6.7% in carbon emissions, which was higher than the national average and showed that its low-carbon development was moving slowly. Shandong and Henan Provinces constantly made up over half of the total emissions but their average annual growth rates dropped to 2.3% and 1.1%, respectively. This implied that they had achieved some success in the low-carbon transformation in the industry.

The analysis of carbon emission figures from nine provinces and autonomous regions in the Yellow River Basin clarified the spatial and temporal changes in emission patterns among the upper, middle, and lower parts of the Basin. The differences might have been ascribed to alterations in national policies during distinct development periods, diverse ecological development traits, different industrial make-ups, and uneven levels of economic progress. These altogether signified a complicated interaction of elements and accomplished the region's carbon reduction and neutrality objectives dependent on separating economic growth from the traditional reliance on fossil fuels by decreasing energy consumption. Regional plans suited to the ecological and industrial features of each place had to be carried out to protect the environment, while improving the industrial structure included boosting low-carbon areas like ecological industries and green agriculture. Other essential improvements encompassed modifying energy composition, developing clean energy technologies, and cutting down on traditional energy sources such as coal. These actions needed to be in line with the national spirit of sustainable development which stressed innovation, coordination, green development, openness, and shared prosperity to create a balanced situation among energy use, economic operations, and carbon emissions in the Yellow River Basin.

Table 1. Data of carbon emissions by province and region within the Yellow River Basin

Year	Qinghai	Gansu	Ningxia	Inner Mongolia	Sichuan	Shaanxi	Shanxi	Henan	Shandong
2006	24.358	89.405	58.832	290.674	189.775	128.683	320.115	378.967	605.513
2007	25.969	97.787	66.500	340.002	208.484	148.118	344.941	426.239	662.955
2008	31.574	103.347	75.390	411.425	230.142	165.362	370.906	435.562	697.730
2009	31.681	95.423	76.899	432.716	236.950	172.869	367.423	416.219	677.008
2010	32.011	127.587	93.782	474.826	265.586	225.257	432.350	513.612	752.651
2011	33.809	132.297	135.585	597.665	266.445	228.917	462.897	515.903	786.967
2012	44.908	154.249	136.586	635.165	338.287	268.677	501.495	529.246	872.906
2013	48.149	160.899	144.108	590.556	351.769	273.347	508.758	492.902	794.206
2014	48.753	164.619	144.573	600.610	351.480	285.344	498.323	544.977	819.064
2015	51.356	160.085	141.821	601.546	332.377	284.025	461.609	527.431	854.463
2016	56.672	153.935	139.493	606.107	319.098	272.289	474.564	521.472	863.426
2017	53.486	151.959	178.624	656.847	318.841	274.030	508.588	501.856	835.819
2018	51.938	162.990	191.589	723.569	296.313	276.166	541.684	490.678	901.647
2019	51.752	164.488	212.414	794.279	315.163	296.273	564.863	460.631	937.117
2020	47.927	175.870	225.911	839.743	307.553	309.785	583.249	473.657	930.639
2021	56.381	189.453	235.318	843.399	314.902	339.104	613.728	483.738	947.163

Note: Unit: million metric tons.

**Figure 2.** Changes of carbon emissions over time in the upper, middle, and lower reaches of the Yellow River Basin: (a) upstream; (b) midstream; (c) downstream

4.2 Analysis of Decoupling Status

Between 2006 and 2021, the decoupling index (ΔCO_2) and the change in Gross Domestic Product (ΔGDP) for nine provinces and autonomous regions in the Yellow River Basin were calculated using Eq. (1). Subsequently, the decoupling stage for each province and autonomous region was determined according to the classification standards presented in Figure 1; the obtained results were summarized in Tables 2–4. To systematically analyze the phased characteristics of carbon emission decoupling in the Basin and align with the implementation cycles of China’s national low-carbon development policies and five-year plans, the 16-year research period was divided into four equal-length sub-periods: 2006–2009, 2010–2013, 2014–2017, and 2018–2021. This division was based on three core considerations: First, it ensured a consistent sample size of 4 years for each sub-period, thus eliminating statistical bias caused by unequal time intervals in comparative analysis; second, it closely corresponded to the key turning point of China’s national carbon reduction policies, with each sub-period covering a major policy implementation phase; third, it is consistent with the widely adopted phased analysis framework in existing studies on carbon emission decoupling in the Yellow River Basin. Specifically, the 2006–2009 period covered the early stage of the 11th Five-Year Plan (2006–2010), when China first incorporated energy conservation and emission reduction as binding targets into national development planning. The 2010–2013 period spanned the later stage of the 11th Five-Year Plan and the early stage of the 12th Five-Year Plan (2011–2015), marked by the launch of China’s first seven regional carbon emission trading pilots in 2013. The 2014–2017 period covered the later stage of the 12th Five-Year Plan and the early stage of the 13th Five-Year Plan (2016–2020), during which the national carbon emission trading system was officially approved for construction in 2017. The 2018–2021 period spanned the later stage of the 13th Five-Year Plan and the first year of the 14th Five-Year Plan (2021–2025), culminating in China’s announcement of the “Dual Carbon” goals (carbon peak by 2030 and carbon neutrality by 2060) in September 2020. This phased division enabled a more accurate identification of the impact of different environments for policies on the decoupling relationship between economic growth and carbon emissions in the Yellow River Basin. Furthermore, a spatiotemporal evolution map was constructed to visualize the dynamic changes of carbon emission decoupling states across the Yellow River Basin, as shown in Figure 3.

Table 2. Analysis of decoupling between carbon emissions in nine provinces of the Yellow River Basin (Year 2006–2021)

Time Period	Qinghai	Gansu	Ningxia	Inner Mongolia	Sichuan	Shaanxi	Shanxi	Henan	Shandong
2006–2007	0.287	0.437	0.458	0.703	0.405	0.639	0.299	0.525	0.480
2007–2008	0.879	0.384	0.448	1.009	0.500	0.442	0.347	0.111	0.272
2008–2009	0.071	-1.198	0.179	0.375	0.263	0.397	0.899	-0.545	-0.331
2009–2010	0.048	1.631	0.912	0.631	0.565	1.312	0.719	1.292	0.753
2010–2011	0.284	0.167	1.946	1.686	0.015	0.069	0.316	0.028	0.301
2011–2012	2.845	1.387	0.072	0.586	1.977	1.075	1.152	0.258	1.096
2012–2013	0.597	0.374	0.597	-0.797	0.367	0.139	0.556	-0.745	-0.883
2013–2014	0.160	0.276	0.051	0.253	-0.009	0.466	-2.287	1.136	0.432
2014–2015	0.604	-4.700	-0.446	0.024	-1.082	-0.162	3.450	-0.444	0.486
2015–2016	0.842	-0.717	-0.210	0.117	-0.434	-0.645	3.020	-0.132	0.167
2016–2017	-0.614	-0.207	1.863	1.041	-0.006	0.050	0.337	-0.331	-0.442
2017–2018	-0.252	0.694	0.750	1.218	-0.536	0.068	0.640	-0.195	1.365
2018–2019	-0.051	0.121	1.601	1.472	0.788	0.942	0.680	-4.366	0.674
2019–2020	-3.165	2.308	1.146	21.653	-0.524	5.325	0.632	2.805	-0.216
2020–2021	1.415	0.557	0.261	0.019	0.207	0.599	0.185	0.303	0.128

An inquiry into the disconnection of carbon emissions across the Yellow River Basin disclosed that the majority of instances exhibited a tendency of positive disconnection, namely a tiny amount of separation which linked carbon emissions and economic growth. The growth rates of carbon emissions were constantly lower than those of the economy and at the same time, sustainable development and expansive negative decoupling phenomena were found out. It was noteworthy that Qinghai and Sichuan were models of sustainable development, while Inner Mongolia and Ningxia mostly presented expansive negative decoupling because of the enlargement of energy-intensive and heavy-industry areas. Shanxi and Shaanxi were highly reliant on coal and they demonstrated substantial changes in their disconnection situation. In the downstream provinces of Shandong and Henan, disconnection trends mostly switched between sustainable development and weak decoupling, even though they faced some unfavorable circumstances. These regions, being crucial industrial centres, had made visible headway in advancing low-carbon development.

The spatial distribution of carbon emission sequestration within the Yellow River Basin showed great unevenness.

When we analyzed the spatial distribution as shown in Figure 3, along with the statistical results recorded in Tables 3 and 4, it was found that the upper part of the Basin, especially in Qinghai and Sichuan Provinces, always had a marked trend toward sequestration. The implementation of clean-energy technology played a huge part in reducing carbon emissions although there were times when sequestration increased in all provinces. Inner Mongolia had a setback towards higher sequestration four times more often and the upper part of the Basin often displayed an upward trend of connection. This meant that its sequestration performance was not adequate because for a long time, industries like non-ferrous metals and chlor-alkali chemicals depended on coal and natural gas, thus holding back green emission-reduction activities. In the middle part of the Basin, Shanxi Province entered a recessive decoupling status while Shaanxi Province exhibited an enhanced negative disconnect pattern. Over the past 16 years, both provinces generated more connection, indicating negative environmental effects. This area had high carbon emissions mainly because there were many traditional heavy and chemical industries, which caused serious environmental harm and had an adverse impact on the sequestration process. The downstream area was more stable regarding carbon emission sequestration and there were fewer cases of unfavorable situations, as can be seen from Figure 3. Shandong Province mostly stayed in a weak disconnection situation during the evaluation period whereas Henan Province often reached a stable disconnection situation. The relatively good sequestration states in these provinces revealed the positive effect of strong national policy support and the fast growth of the digital industry, which has been very helpful in promoting green development in the lower part of the Yellow River Basin.

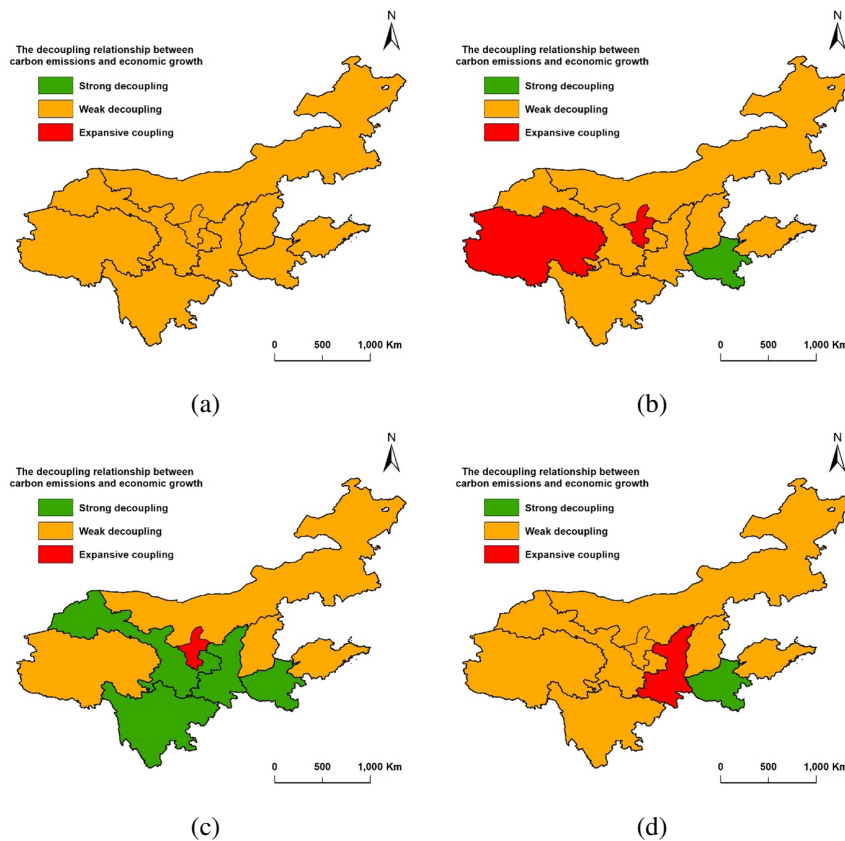


Figure 3. Spatial distribution of carbon emission decoupling states in nine provinces of the Yellow River Basin: (a) 2006–2009; (b) 2010–2013; (c) 2014–2017; (d) 2018–2021

Examining the decoupling scenarios of carbon emissions in the Yellow River Basin, we could observe how they varied over time and figure out the spatial distribution patterns shown in Figure 3 and the exponential data presented in Table 2. Between 2006 and 2009, in most provinces, there was a connection between economic growth and carbon emissions. Meanwhile, many provinces were dominated by a large-scale economic model, which contributed to economic activities throughout the nine Yellow River provinces. From 2010 to 2013, the differences in decoupling became more pronounced. Inner Mongolia, Ningxia, and Gansu had energy and demand disconnections while Qinghai illustrated a positive link between carbon emissions and growth as in Figure 3. This period corresponded to the early stage of China's 12th Five-Year Plan when the economy and society were developing rapidly and the need for energy was increasing, resulting in a substantial growth of carbon emissions. At the same time, weak decoupling was common in the middle and lower parts of the Basin while strong decoupling could only be witnessed in Shandong

and Henan from 2012 to 2013, thus widening the gap with the upper region. Between 2014 and 2021, the policy relaxation could be divided into two sub-periods: 2014–2017 and 2018–2021. The release of the Outline of the Plan for Ecological Protection and High-Quality Development of the Yellow River Basin by the Central Committee of the Communist Party of China and the State Council emphasized the importance of optimizing energy development in line with water resources and ecological carrying capacities. During this time, downstream provinces gradually improved their energy utilization structure and efficiency, rendering the strong decoupling zone larger and weak decoupling a common occurrence. Shanxi located in the middle of the Basin had a reduction and disconnection between 2014 and 2015. Then it turned to weak decoupling after 2016 while Shaanxi kept a favorable disconnection situation. The high-risk situation in the upper part decreased significantly and Ningxia sometimes had expansive negative decoupling. Between 2018 and 2021, the disconnection in the Yellow River Basin changed further and the middle and lower parts tended to be stable. Inner Mongolia, which is a crucial national energy and strategic resource base in the upper area, often faced expansive negative decoupling especially between 2017 and 2020, hence indicating the impact of coal and power supply on the disconnection pattern. However, between 2020 and 2021, it returned to a weak decoupling state apart from the active development of new energy sources and the construction of a modern, clean, and low-carbon energy system that offered the Yellow River Basin a great route for decoupling.

Table 3. Decoupling status of carbon emissions in provinces of the upper Yellow River Basin

Time Period	Qinghai	Gansu	Ningxia	Inner Mongolia	Sichuan
2006–2007	Weak decoupling	Weak decoupling	Weak decoupling	Weak decoupling	Weak decoupling
2007–2008	Expansive coupling	Weak decoupling	Weak decoupling	Expansive coupling	Weak decoupling
2008–2009	Weak decoupling	Strong decoupling	Weak decoupling	Weak decoupling	Weak decoupling
2009–2010	Weak decoupling	Expansive negative decoupling	Expansive coupling	Weak decoupling	Weak decoupling
2010–2011	Weak decoupling	Weak decoupling	Expansive negative decoupling	Expansive negative decoupling	Weak decoupling
2011–2012	Expansive negative decoupling	Expansive negative decoupling	Weak decoupling	Weak decoupling	Expansive negative decoupling
2012–2013	Weak decoupling	Weak decoupling	Weak decoupling	Strong decoupling	Weak decoupling
2013–2014	Weak decoupling	Weak decoupling	Weak decoupling	Weak decoupling	Strong decoupling
2014–2015	Weak decoupling	Strong decoupling	Strong decoupling	Weak decoupling	Strong decoupling
2015–2016	Expansive coupling	Strong decoupling	Strong decoupling	Weak decoupling	Strong decoupling
2016–2017	Strong decoupling	Strong decoupling	Expansive negative decoupling	Expansive coupling	Strong decoupling
2017–2018	Strong decoupling	Weak decoupling	Weak decoupling	Expansive negative decoupling	Strong decoupling
2018–2019	Strong decoupling	Weak decoupling	Expansive coupling	Expansive negative decoupling	Weak decoupling
2019–2020	Strong decoupling	Expansive negative decoupling	Weak decoupling	Expansive negative decoupling	Strong decoupling
2020–2021	Expansive negative decoupling	Weak decoupling	Weak decoupling	Weak decoupling	Weak decoupling

Table 4. Decoupling analysis of carbon emissions in the middle and lower reaches of the Yellow River Basin

Time Period	Midstream		Downstream	
	Shaanxi	Shanxi	Henan	Shandong
2006–2007	Weak decoupling	Weak decoupling	Weak decoupling	Weak decoupling
2007–2008	Weak decoupling	Weak decoupling	Weak decoupling	Weak decoupling
2008–2009	Weak decoupling	Recessive coupling	Strong decoupling	Strong decoupling
2009–2010	Expansive negative decoupling	Weak decoupling	Expansive negative decoupling	Weak decoupling
2010–2011	Weak decoupling	Weak decoupling	Weak decoupling	Weak decoupling
2011–2012	Expansive coupling	Expansive coupling	Weak decoupling	Expansive coupling
2012–2013	Weak decoupling	Weak decoupling	Strong decoupling	Strong decoupling
2013–2014	Weak decoupling	Strong decoupling	Expansive coupling	Weak decoupling
2014–2015	Strong decoupling	Recessive decoupling	Strong decoupling	Weak decoupling
2015–2016	Strong decoupling	Expansive negative decoupling	Strong decoupling	Weak decoupling
2016–2017	Weak decoupling	Weak decoupling	Strong decoupling	Strong decoupling
2017–2018	Weak decoupling	Weak decoupling	Strong decoupling	Expansive negative decoupling
2018–2019	Expansive coupling	Weak decoupling	Strong decoupling	Weak decoupling
2019–2020	Expansive negative decoupling	Weak decoupling	Expansive negative decoupling	Strong decoupling
2020–2021	Weak decoupling	Weak decoupling	Weak decoupling	Weak decoupling

4.3 Model Checking and Grey Prediction Results

Time-series data regarding carbon emissions from 2006 to 2021 was utilized. It was gathered from the nine provinces in the Yellow River Basin, and a grey forecasting model was then carried out to predict future emissions. After that, the expected carbon emissions for these provinces from 2022 to 2026 were calculated by means of an Excel spreadsheet and the outcomes are shown in Table 5.

Table 5. Predicted carbon emissions from nine provinces in the Yellow River Basin

Time Period	Qinghai	Gansu	Ningxia	Inner Mongolia	Sichuan	Shaanxi	Shanxi	Henan	Shandong
2022	61.855	195.836	262.810	904.652	347.887	352.231	625.466	509.259	986.955
2023	64.576	203.398	284.582	953.223	354.985	367.456	646.943	511.871	1010.368
2024	67.416	211.253	308.158	1004.402	362.227	383.339	669.157	514.497	1034.337
2025	70.381	219.411	333.688	1058.328	369.617	399.908	692.134	517.136	1058.875
2026	73.477	227.885	361.332	1115.149	377.159	417.193	715.900	519.789	1083.994

The precision of the GM(1,1) model was evaluated through residual analysis and posterior error test. To comprehensively verify the applicability of the model in different regions of the Yellow River Basin, three representative provinces covering the upper, middle, and lower reaches were selected for accuracy testing: Inner Mongolia Autonomous Region, Shanxi Province, and Shandong Province.

(1) Residual analysis:

To figure out the difference between the forecasted and real-world value $\hat{x}^{(0)}(i)$, the residuals e_i and relative errors ε_i were computed using the following formula:

$$e_i = x^{(0)}(i) - \hat{x}^{(0)}(i), i = 2, 3, \dots, 16 \quad (5)$$

$$\varepsilon_i = \frac{e_i}{x^{(0)}(i)} \quad (6)$$

The average relative error $\bar{\varepsilon}_i$ was then calculated as:

$$\bar{\varepsilon}_i = \frac{1}{15} \sum_{i=2}^{16} \varepsilon_i \quad (7)$$

According to the standard precision grading system for grey prediction models, the average relative error was divided into four levels: first-level precision $\bar{\varepsilon}_i < 0.05$, second-level precision $0.05 \leq \bar{\varepsilon}_i < 0.1$, third-level precision $0.1 \leq \bar{\varepsilon}_i < 0.2$ and fourth-level precision $\bar{\varepsilon}_i \geq 0.2$. The calculation results showed that the average relative errors of Inner Mongolia Autonomous Region, Shanxi Province, and Shandong Province were 0.0617, 0.0501, and 0.027 respectively. Specifically, Inner Mongolia Autonomous Region was $\bar{\varepsilon}_i \approx 0.0617$, falling within the range of second-level precision. Shanxi Province was $\bar{\varepsilon}_i \approx 0.0501$, falling within the range of second-level precision. Shandong Province was $\bar{\varepsilon}_i \approx 0.027$, falling within the range of first-level precision.

(2) Posterior error test:

According to the formula, the posterior error test was conducted using the mean square variance ratio C and small error probability P , calculated as follows:

$$S_1 = \sqrt{\frac{1}{n-1} \sum_{k=1}^n [\hat{x}^{(0)}(i) - \overline{x^{(0)}}]^2}, \overline{x^{(0)}} = \frac{1}{n} \sum_{k=1}^n \hat{x}^{(0)}(i) \quad (8)$$

$$S_2 = \sqrt{\frac{1}{n-1} \sum_{k=1}^n [e_i - \bar{e}_i]^2}, \bar{e}_i = \frac{1}{n} \sum_{k=1}^n e_i \quad (9)$$

$$C = \frac{S_2}{S_1}, p = P\{|e_i - \bar{e}| < 0.6745S_1\} \quad (10)$$

According to the grey system theory, precision grading standard $C < 0.35$ and $p > 0.95$ belong to the first level; $0.35 \leq C < 0.5$ and $0.80 < p \leq 0.95$ belong to the second-level precision.

For Inner Mongolia Autonomous Region, $C \approx 0.276 < 0.35$, $p = 1 > 0.95$, thus attaining the first-level precision. The result of Shanxi Province was $C \approx 0.333 < 0.35$, $p = 1 > 0.95$, reaching the first-level precision. As regards Shandong Province, $C \approx 0.308 < 0.35$, $p = 0.9375 > 0.8$, hence meeting the second-level precision.

The above experiments demonstrated that although the average relative errors of Inner Mongolia and Shanxi were at the second-level precision due to the inherent volatility of carbon emission systems, the GM(1,1) model demonstrated excellent overall stability and reliability in all three representative regions of the Yellow River Basin, as evidenced by the high-level posterior error test results. The model fully met the requirements for forecasting the medium-term development trend of carbon emissions in the Basin from 2022 to 2026. The comparison between the observed and predicted carbon emissions for the three provinces from 2006 to 2021 is shown in Table 6.

Table 6. Comparison of actual and predicted carbon emissions in representative provinces of the Yellow River Basin (Year 2006–2021) based on GM(1,1) model

Year	Inner Mongolia			Shanxi			Shandong		
	Actual	Predicted	Relative Error	Actual	Predicted	Relative Error	Actual	Predicted	Relative Error
2006	290.674	290.670	0.000	320.115	320.115	0.000	605.513	605.513	0.000
2007	340.002	412.850	0.045	344.941	376.939	31.998	662.955	694.323	31.368
2008	411.425	435.010	0.018	370.906	389.882	18.976	697.730	710.794	13.064
2009	432.716	458.370	0.070	367.423	403.269	35.846	677.008	727.656	50.648
2010	474.826	482.980	-0.010	432.350	417.117	-15.233	752.651	744.918	-7.733
2011	597.665	508.910	-0.032	462.897	431.439	-31.458	786.967	762.590	-24.377
2012	635.165	536.230	-0.118	501.495	446.254	-55.241	872.906	780.681	-92.225
2013	590.556	565.020	0.006	508.758	461.577	-47.181	794.206	799.200	4.994
2014	600.610	595.360	-0.001	498.323	477.426	-20.897	819.064	818.160	-0.904
2015	601.546	627.320	-0.020	461.609	493.820	32.211	854.463	837.569	-16.894
2016	606.107	661.010	-0.007	474.564	510.776	36.212	863.426	857.438	-5.988
2017	656.847	696.490	0.048	508.588	528.315	19.727	835.819	877.779	41.960
2018	723.569	733.890	-0.003	541.684	546.456	4.772	901.647	898.603	-3.044
2019	794.279	773.290	-0.019	564.863	565.219	0.356	937.117	919.920	-17.197
2020	839.743	814.810	0.012	583.249	584.628	1.379	930.639	941.743	11.104
2021	843.399	858.560	0.018	613.728	604.702	-9.026	947.163	964.084	16.921

5 Discussion

This study tactfully employed the Tapio decoupling model and the GM(1,1) grey prediction model to systematically characterize the spatiotemporal evolution of carbon emissions across the nine provinces in the Yellow River Basin from 2006 to 2021. It identified the overall pattern of carbon emissions in the Basin, marked by “rapid expansion in the middle and upper reaches and dominance by the lower reaches in total volume”. Furthermore, the study revealed the gradient differences in the decoupling relationship between economic growth and carbon emissions, specifically “deterioration in the upper reaches, fluctuation in the middle reaches, and improvement in the lower reaches”. It provided a scientific prediction of carbon emission trends from 2022 to 2026.

This pronounced spatial heterogeneity was essentially the outcome of the long-term interplay of resource endowments, industrial division of labor, and policy orientations within the Basin. As national core energy bases, Inner Mongolia and Ningxia in the upper reaches have developed an industrial structure centred on coal-fired power and coal chemical industries by leveraging their abundant coal resources. The sustained expansion of energy-intensive heavy industries has pushed their carbon emission growth rates far above the Basin average. They have frequently experienced expansive negative decoupling under the influence of the national coal power supply guarantee policy. Shanxi and Shaanxi in the middle remains highly dependent on the coal industry, and their decoupling status reveals significant instability in response to coal market cycles and adjustments to overcapacity reduction policies. Shandong and Henan in the lower reaches embarked on industrial transformation earlier, with the share of energy-intensive industries declining steadily and the tertiary sector and high-tech manufacturing emerging as the main engines of economic growth. As a result, their carbon emission growth rates have slowed down markedly, with weak decoupling being the dominant state and the frequency of strong decoupling increasing continuously.

Unlike most existing studies that focused on single-dimensional carbon emission accounting or trend prediction, this study organically integrated spatiotemporal heterogeneity analysis, economic decoupling evaluation, and trend prediction, thus better reflecting the qualitative differences in low-carbon development across the Basin. In terms of methodological adaptability, the GM(1,1) model effectively addressed the challenge of small-sample prediction arising from incomplete statistical data in some provinces of the Yellow River Basin. Moreover, the prediction results were effectively aligned with the historical evolution of decoupling, rendering them more policy-targeted.

Nevertheless, this study has several limitations. The contribution of each driving factor has not been precisely quantified through quantitative models. Limited by data availability, it does not cover the latest impacts of the deepening of the “Dual Carbon” policies and post-pandemic economic recovery since 2022. In addition, the GM(1,1) model has limited capacity to capture external shocks such as abrupt policy changes. The production-based carbon accounting also fails to consider inter-provincial carbon emission transfers, which may lead to biases in the allocation of emission reduction responsibilities.

6 Conclusions

The present research examined carbon emission figures from 2006 to 2021 covering nine provinces and autonomous areas in the Yellow River Basin. It utilized the Tapio decoupling framework along with the GM(1,1) grey forecasting model to map out the course of carbon emissions and predict future tendencies. Several crucial discoveries were derived: (1) Carbon emissions in the Yellow River Basin demonstrated obvious periodicity, growing from around 2,086 million metric tons to 4,023 million metric tons during the research period, with an approximate annual growth rate of 4.2%. Regional differences were clear, with faster growth in the mid- and upper-reaches and higher emissions in the lower-reach areas. (2) The Basin mainly witnessed weak decoupling of carbon emissions which varied significantly in space and in the upper regions, especially in Inner Mongolia and Ningxia. Expansive negative decoupling often happened where carbon emissions increased more than the economy while Qinghai and Sichuan sometimes showed strong decoupling and the upper area had bad decoupling results. In the middle-reach, Shanxi and Shaanxi experienced sharp changes, showing both reduction and expansive negative decoupling. Shandong in the lower-reach mostly displayed weak decoupling; Henan saw enhanced strong decoupling to represent the best decoupling situation. (3) The carbon emission paths of the nine provinces in the Yellow River Basin were divided into three stages: rapid growth, increased differentiation, and policy adjustment. As national economic goals and policies changed, different decoupling levels were witnessed in various river basins but driven by the “Dual Carbon” policy aiming at the decoupling trend of carbon emissions in these Basins, which was heading towards an ideal state. (4) The examination of carbon emission routes and decoupling states in the nine provinces of the Yellow River Basin clarified connections with national policies, ecological development traits, industrial structures, and the levels of economic development at certain times, which were considered to be the results of multiple converging factors.

A detailed study was carried out to examine how carbon emissions developed and varied in the nine provinces surrounding the Yellow River. After that, a comprehensive evaluation was conducted to assess whether carbon emissions were becoming less dependent on economic growth. Then, a GM(1,1) grey prediction model was created to direct the future path of carbon emissions for every single area. The results of the model suggested different

tendencies in carbon emissions among the provinces and nearby places along the Yellow River. Overall, the model demonstrated that there would be a surge of carbon emissions in these regions and provinces between 2022 and 2026.

Based on the inferences mentioned before, some suggestions were put forward to help achieve continuous carbon emission reduction in the Yellow River Basin and promote environmentally sustainable development:

(1) To facilitate the industrial sectors in the Yellow River Basin to shift towards low-carbon practices, we worked out specific strategies matching the different industrial make-ups of the upper, middle, and lower parts of the Basin. Specifically speaking, the upper-reach regions had to strictly limit the growth of energy-intensive industries, take advantage of their closeness to clean energy sources, and promote the coordinated development of wind and solar power generation, together with ecological protection. In the middle-reach areas, the focus was on guiding the coal industry towards more environmentally-friendly practices, providing technological improvements to cut down carbon emissions in proportion to the output value, and aligning the functions of energy-producing bases with low-carbon goals. Meanwhile, the lower-reach regions were required to speed up industrial restructuring, increase energy efficiency in high-carbon fields, incorporate low-carbon technologies, and strengthen the sustainability of low-carbon industrial development. Moreover, it was crucial to establish a cross-regional industrial cooperation framework to boost the connection between low-carbon value chains of upstream and downstream industries so as to reduce the likelihood of carbon-heavy industries relocating to other places.

(2) To make regional collaborative emission reduction strategies and governance frameworks more effective, an integrated carbon emission collaborative governance network had to be developed for the Yellow River Basin. It would be easier to standardize emission reduction criteria and establish a unified accounting system, to ensure that the emission reduction goals and action plans of different provinces were aligned. It was crucial for the provinces to maintain coherence by linking the clean energy-related emission reduction in upstream areas to the requirements of downstream regions. Regional carbon emission trading could be promoted to equalize regional interests via market-based mechanisms. More investment should have been put into scientific and technological progress to support the research and deployment of low-carbon technologies throughout the regions, while emphasis should have been placed on critical technological breakthroughs like innovative energy storage solutions and reduction of carbon footprints in existing industries. Moreover, promoting the exchange of emission reduction knowledge and dealing with the obstacles to regional cooperation were essential for achieving efficient emission reduction.

(3) To construct a superior energy structure across the whole river basin, an all-round framework was preferred to help shift towards a cleaner and low-carbon energy pattern. In the upper part of the river basin, wider use of renewable energy like wind and solar power had to be promoted when improving energy storage and transmission systems so that clean energy could be distributed efficiently. For the middle part, focus should be put on making use of coal as clean and efficient as possible, encouraging combination of the coal and power industries, and lowering carbon emissions from conventional energy sources by utilizing carbon capture and storage technologies. In the lower part of the river basin, the primary goal of reducing carbon emissions could be achieved by improving energy consumption system, increasing the proportion of clean energy in the total energy supply, promoting cooperation among different regions regarding clean energy resources, and curtailing our dependence on fossil fuels.

Author Contributions

Conceptualization, H.H.; methodology, H.H. and Y.W.H.; software, Y.W.H. and B.W.Z.; validation, Y.J.H., J.W., and W.X.D.; formal analysis, B.W.Z. and B.J.Z.; investigation, W.X.D. and S.H.C.; resources, S.H.C. and B.J.Z.; data curation, Y.W.H. and B.W.Z.; writing—original draft preparation, Y.W.H.; writing—review and editing, H.H. and A.X.L.; visualization, B.W.Z. and S.H.C.; supervision, H.H.; project administration, H.H.; funding acquisition, H.H. and A.X.L. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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